

Shuaib Iqbal

MACHINE LEARNING ENGINEER | DATA SCIENTIST | AI ENGINEER | DJANGO BACKEND DEVELOPER

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Summary

Data Scientist and Machine Learning Engineer with expertise in machine learning, artificial intelligence, data science, and natural language processing (NLP). Proficient in building scalable data pipelines on Linux, automating workflows with Python, and deploying AI models using Docker.

Additionally experienced in Django-based backend development, REST API design, and database-driven web applications. Skilled in integrating ML systems into scalable web architectures and deploying data-driven backend services using Django and PostgreSQL.

Work Experience

Machine Learning Engineer

AiSight.ai, Lahore

Oct 2022 – Present

- Spearheaded AI-driven Revenue Growth Management (RGM) initiatives for FMCG leaders (EBM, PMI, RB-PK, Unilever), driving double-digit growth.
- Built predictive models and analytical dashboards for Precision Selling and route optimization.
- Developed “White Space Identification” and “Route Performance” algorithms improving route productivity.
- Engineered time-windowed stockout forecasting models and alerting systems reducing out-of-stock events by 20%.
- Designed robust data pipelines (Python, SQL) to unify POS and census data.
- Deployed ML/DL models for actionable commercial outcomes.
- Built NLP-powered SQL generation tool using LLMs.
- Containerized ML workflows using Docker for SaaS deployment.
- Developed backend services using Django/FastAPI to expose ML insights via REST APIs.
- Optimized PostgreSQL queries and contributed to backend architecture of internal production web systems.

Associate Machine Learning Engineer

Sense Digital, Islamabad

May 2022 – Oct 2022

- Improved HVAC forecasting accuracy by 25% using ARIMA, LSTM, and XGBoost.
- Deployed anomaly detection systems reducing energy wastage.
- Integrated predictive ML models into IoT systems.
- Streamlined sensor data processing pipelines.
- Built interactive dashboards (Streamlit, Plotly).
- Automated end-to-end reporting through REST APIs.
- Designed structured databases and backend services for handling time-series IoT datasets.

- Delivered sales forecasting models (Prophet, LSTM, XGBoost).
- Conducted RFM segmentation and customer analytics.
- Engineered features and evaluated models using MAE, RMSE, and R^2 .
- Created interactive business dashboards.
- Built backend scripts and reporting services for dashboard applications.
- Worked with relational databases to structure and serve processed datasets.

Lecturer – Computer Science

Nov 2018 – May 2019

Eclat College for Girls, Rawalpindi

- Taught programming fundamentals, data structures, and software engineering.
- Guided final-year projects focused on practical software solutions.

Education and Training

MS Computer Engineering

2018 – 2021 | CGPA: 3.50/4.00

NUST, Islamabad

Focus: Machine Learning, Pattern Recognition, Image Processing, DSP**Thesis:** Predicting Residential Energy Consumption via Encoder-Decoder CNN-LSTM**B.Sc. Computer Systems Engineering**

2014 – 2018 | CGPA: 3.62/4.00

IUB, Bahawalpur

Focus: AI and Robotics, Image Processing, Data Mining, Probability**Thesis:** Pattern Formation of Swarm Robots

Technical Skills

- **Languages/Frameworks:** Python (Pandas, NumPy, TensorFlow, Keras, PyTorch, Scikit-learn, SciPy), Django, Django REST Framework, FastAPI, Streamlit
- **ML & NLP:** Classical ML, Deep Learning, Transformers, LangChain, RAG, HuggingFace, OpenAI API
- **Databases:** SQL, PostgreSQL, SQLAlchemy, ETL Pipelines
- **Backend:** REST APIs, Authentication, Django ORM, Database Design
- **DevOps:** Git, Docker, Linux
- **Tools:** VS Code, PyCharm, Jupyter, Google Colab

Projects and Research

- Time-Series Prediction of Cryptocurrency Market using ML
- Forecasting Stock Market using Encoder-Decoder ConvLSTM
- Multilevel Feedback Queue Scheduling with Dynamic Quantum
- Census Data Prediction
- Robotic Arm with Inverse Kinematics
- CNN for Handwritten Digit Recognition
- Database Management for Alpha Electronics (Oracle 11g)

Selected Projects

- **AI-Powered Resume Assistant**

Streamlit · LangChain · Ollama — Engineered a fully offline, privacy-first LLM-based assistant leveraging local models for intelligent resume content generation, structural optimization, and contextual rewriting.

- **Context-Aware AI Chatbot**

Streamlit · LangChain · Ollama — Designed and implemented a memory-enabled conversational AI system supporting contextual recall and document-grounded Q&A for secure internal deployments.

- **YouTube Comment Sentiment Analyzer**

YouTube API · NLP · Sentiment Analysis — Developed a data pipeline to extract, preprocess, and analyze user engagement data, enabling sentiment classification and audience behavior insights.

- **Automated Web Scraping and Data Extraction Pipelines**

Python · Selenium · BeautifulSoup — Built robust, scalable scraping systems for e-commerce platforms, blogs, and geo-business listings, implementing CAPTCHA handling, structured parsing, and automated data cleaning workflows.

- **Django Web Applications Suite**

Django · ORM · REST API Integration · SQLite — Designed and developed multiple modular web applications using Django's MTV architecture, implementing secure CRUD operations with CSRF protection, relational database modeling via Django ORM, REST API integration with JSON parsing and exception handling, dynamic template rendering using DTL with template inheritance, media upload handling, persistent search tracking, and structured admin-based content management.

Certifications

- **Python and Data Science:**

- Python Basics, Programming for Everybody – University of Michigan (Coursera)
- Introduction to Python for Data Science – IBM (Coursera)
- Data Analysis with Pandas – IBM (Coursera)
- Data Visualization with Plotly – Coursera

- **Artificial Intelligence & Machine Learning:**

- Introduction to AI & Prepare for AI Engineering – Microsoft (Learn)
- Microsoft Azure AI Fundamentals: Text, Speech, Vision, Language Understanding, Question Answering, Machine Learning, Knowledge Mining

- **DevOps Tools:**

- Docker for Beginners – Udemy Labs